SRI (k-inflation): LO

The spectra are evaluated at $-k_*\eta_* = 1$, through relative amplitude order $p = 0$. The Hubble-flow parameters are denoted by ϵ_j and $M_{\rm Pl}$ is the reduced Planck mass. Coefficients are generated numerically from the normalized mode equations, using the iterative algebra. The convention for the spectral derivatives is

$$\alpha_n^A = \frac{d^{n+1} \ln P_A}{d(\ln k)^{n+1}}, \qquad \alpha_0^s = n_s - 1, \qquad \alpha_0^t = n_t, \qquad r = P_t/P_s.$$

For k-inflation, $c_t = 1$. The scalar amplitude includes transport from the acoustic reference to the common tensor reference. The sound speed is $c_{s*} > 0$, with

$$\alpha_n^{c_s} = \frac{d^{n+1} \ln c_s}{d(\ln k)^{n+1}}.$$

1 Spectra

1.1 Scalar spectrum

$$P_s(k_*) = \frac{H_*^2}{8\pi^2 M_{\rm Pl}^2 \epsilon_{1*} c_{s*}} \left[\sum_{d=0}^{0} F_{s,d} + O(\mathrm{SR}^1) \right].$$

Relative order 0.

$$F_{s,0} = 1$$

1.2 Tensor spectrum

$$P_t(k_*) = \frac{2H_*^2}{\pi^2 M_{\rm Pl}^2} \left[\sum_{d=0}^{0} F_{t,d} + O(\mathrm{SR}^1) \right].$$

Relative order 0.

$$F_{t,0} = 1$$

2 Consistency relations

2.1 $n = 0$

For $p = 0$, the absolute cutoff is $W = 1$:

$$\alpha_0^t = \sum_{d=1}^{1} R_{0,d} + O(\mathrm{SR}^2).$$

Absolute order 1.

$$R_{0,1} = -\frac{r}{8c_s}$$

2.2 $n = 1$

For $p = 0$, the absolute cutoff is $W = 2$:

$$\alpha_1^t = \sum_{d=2}^{2} R_{1,d} + O(\mathrm{SR}^3).$$

Absolute order 2.

$$R_{1,2} = \frac{r^2}{64c_s^2} + \frac{\alpha_0^{c_s} r}{8c_s} + \frac{\alpha_0^s r}{8c_s}$$

2.3 $n = 2$

For $p = 0$, the absolute cutoff is $W = 3$:

$$\alpha_2^t = \sum_{d=3}^{3} R_{2,d} + O(\mathrm{SR}^4).$$

Absolute order 3.

$$\begin{aligned} R_{2,3} = &-\frac{r^3}{256c_s^3} - \frac{3\alpha_0^{c_s} r^2}{64c_s^2} \\ &- \frac{3\alpha_0^s r^2}{64c_s^2} - \frac{\left(\alpha_0^{c_s}\right)^2 r}{8c_s} \\ &- \frac{\left(\alpha_0^s\right)^2 r}{8c_s} + \frac{\alpha_1^{c_s} r}{8c_s} \\ &+ \frac{\alpha_1^s r}{8c_s} - \frac{\alpha_0^{c_s}\alpha_0^s r}{4c_s} \end{aligned}$$

2.4 $n = 3$

For $p = 0$, the absolute cutoff is $W = 4$:

$$\alpha_3^t = \sum_{d=4}^{4} R_{3,d} + O(\mathrm{SR}^5).$$

Absolute order 4.

$$\begin{aligned} R_{3,4} = &\frac{3r^4}{2048c_s^4} - \frac{\alpha_1^{c_s} r^2}{16c_s^2} \\ &- \frac{\alpha_1^s r^2}{16c_s^2} + \frac{\alpha_2^{c_s} r}{8c_s} \\ &+ \frac{\alpha_2^s r}{8c_s} + \frac{\left(\alpha_0^{c_s}\right)^3 r}{8c_s} \\ &+ \frac{\left(\alpha_0^s\right)^3 r}{8c_s} + \frac{3\alpha_0^{c_s} r^3}{128c_s^3} \\ &+ \frac{3\alpha_0^s r^3}{128c_s^3} + \frac{7\left(\alpha_0^{c_s}\right)^2 r^2}{64c_s^2} \\ &+ \frac{7\left(\alpha_0^s\right)^2 r^2}{64c_s^2} - \frac{3\alpha_0^{c_s}\alpha_1^{c_s} r}{8c_s} \\ &- \frac{3\alpha_0^{c_s}\alpha_1^s r}{8c_s} - \frac{3\alpha_0^s\alpha_1^{c_s} r}{8c_s} \\ &- \frac{3\alpha_0^s\alpha_1^s r}{8c_s} + \frac{3\alpha_0^{c_s}\left(\alpha_0^s\right)^2 r}{8c_s} \\ &+ \frac{3\left(\alpha_0^{c_s}\right)^2\alpha_0^s r}{8c_s} + \frac{7\alpha_0^{c_s}\alpha_0^s r^2}{32c_s^2} \end{aligned}$$